

XICHONG ZHANG

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Education

University of Science and Technology of China

Master of Engineering in Computer Science

Sept 2022 – Jul 2025

Hefei, China

University of Science and Technology of China

Bachelor of Science in Physics

Sept 2018 – Jul 2022

Hefei, China

Research Interests

My research interests broadly lie in enhancing intelligence in large language models (LLMs), particularly through advanced techniques such as test-time scaling and agentic reasoning. I have explored diverse research topics, ranging from data valuation, client selection, and edge caching decisions, aiming to improve resource utilization efficiency and service quality in distributed systems.

Research Experience

Research Intern

Department of Computer Science, PKU

May 2025– Present

Peking, China

- **Maximizing throughput with latency guarantee for LLM serving**

- Investigated GPU resource efficiency challenges in LLM inference. The conventional prefill-decode (PD) collocation paradigm suffers from poor latency performance, while the PD disaggregation paradigm reduces interference between prefill and decoding stages but leads to suboptimal resource utilization. My research focuses on optimizing GPU harvesting strategies under heterogeneous SLOs and dynamic workloads for both PD-disaggregated and PD-allocated systems.
- In progress.

Graduate Researcher

Department of Computer Science, USTC

Sept 2022– Present

Hefei, China

- **Large-scale mobile edge caching problem considering economic interplay**

- Introduced a large-scale edge caching game model jointly considering content caching, trading, and sharing in the content delivery network. Devised a mean field algorithm to approximate the collective impact of edge servers in order to reduce the computational complexity of obtaining the game equilibrium. Both theoretical analysis and experimental evaluation prove the effectiveness of the solution.
- One paper accepted by ICDE 2024, another first-author manuscript currently under review at IEEE/ACM ToN.

- **Data-driven federated learning using drones**

- Proposed a data-driven federated learning (FL) paradigm with the abstraction of the independent data layer. Constructed data quality metrics and developed an algorithm based on Bayesian inference and the bandit model to achieve optimal scheduling on distributed unknown data. The algorithm achieved better performance on theoretical and real-world datasets.
- First author paper accepted by ICDE 2025.

Undergraduate Researcher

Imaging Quantum Materials Lab, USTC

Sept 2020– Jan 2022

Hefei, China

- **Imaging correlated materials with Scanning Tunneling Microscopy**

- Designed negative-feedback circuits and developed an automated equipment for high-quality Scanning Tunneling Microscopy (STM) tip fabrication. Investigated the interplay between charge density wave and superconductivity in $2H-NbSe_2$ via low-temperature STM.
- Led a 3-member team to win the First Prize in the USTC Physics Research Experiment Competition.

Publications

- **Joint Mobile Edge Caching and Pricing: A Mean-Field Game Approach**

Yin Xu (advisor), **Xichong Zhang**, Mingjun Xiao, Jie Wu, An Liu, and Sheng Zhang
International Conference on Data Engineering (ICDE), 2024

- **Online Federated Learning on Distributed Unknown Data Using UAVs**

Xichong Zhang, Haotian Xu, Yin Xu, Mingjun Xiao, Jie Wu, and Jinrui Zhou
International Conference on Data Engineering (ICDE), 2025

- **An Adaptive Mean-Field Game Framework for Mobile Edge Caching and Pricing**
Xichong Zhang, Yin Xu, Mingjun Xiao, Jie Wu, An Liu, and Sheng Zhang
Submitted to IEEE/ACM Transactions on Networking
- **Multifunctional PVA/PNIPAM conductive hydrogel sensors enabled human-machine interaction intelligent rehabilitation training**
Yanlong Zhao, **Xichong Zhang**, Yilin Hao, Yinghe Zhao, Peng Ding, Wei Zhai, Kun Dai, Guoqiang Zheng, Chuntai Liu, Changyu Shen
Advanced Composites and Hybrid Materials (SCI Q1, IF 21.8)
- **Eco-Friendly Multifunctional Hydrogel Sensors Enabled Sustainable and Accurate Human-Machine Interaction System**
Yanlong Zhao, Rui Wu, Yilin Hao, Yi Zhao, **Xichong Zhang**, Hui Liu, Wei Zhai, Kun Dai, Caofeng Pan, Chuntai Liu, Changyu Shen
Advanced Materials (Top-tier journal in materials science, SCI Q1, IF 26.8)
- **Fuzzy Knowledge-Driven Human-in-the-Loop Learning for Autonomous Driving**
Yiming Zeng, Yin Xu, **Xichong Zhang**, Xiao Huang, Mingjun Xiao
International Joint Conference on Neural Networks (IJCNN), 2025
- **SFedKD: Sequential Federated Learning with Discrepancy-Aware Multi-Teacher Knowledge Distillation**
Haotian Xu, Jinrui Zhou, **Xichong Zhang**, Mingjun Xiao
Under Review
- **CoSine: Enhancing LLM Serving via Collaborative and Decoupled Speculative Inference**
Luyao Gao, Jianchun Liu, Hongli Xu, **Xichong Zhang**, Yunming Liao, Liusheng Huang
Under Review

Selected Honors and Awards

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| • Outstanding Graduate Award, USTC | 2025 |
| • First Class Academic Scholarship, USTC | 2022 |
| • First Prize in College Physics Experiment IV, USTC | 2021 |
| • National Encouragement Scholarship | 2019, 2020 |

Teaching Experience

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| • Teaching Assistant, 011127, Data Structure, USTC | Fall 2022 |
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Skills

- **Languages:** Mandarin (native), English (intermediate)
- **Programming Languages:** Python, C/C++, MATLAB, Bash, Latex
- **Deep Learning Frameworks:** PyTorch, HuggingFace Transformers